

ANDRÉS VIRGÜEZ

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PROFILE

Data Scientist – ML Engineer with 10+ years of experience designing, deploying, and scaling data-driven products across multiple industries. Specializes in end-to-end ML systems, including feature engineering, model development, evaluation, deployment, monitoring, and lifecycle management. Hands-on expertise with Python, SQL, scikit-learn, PyTorch, TensorFlow, MLflow, Docker and cloud platforms (AWS / Azure / GCP), as well as LLMs and RAG pipelines. Adept at translating complex business problems into production-grade solutions, leading cross-functional teams, and aligning technical strategy with organizational priorities.

EXPERIENCE

Senior Data Scientist Oct 2024 – Oct 2025 | Remote

BLD.AI (Consulting for BP)

Developed and deployed end-to-end machine learning and statistical modules (clustering, forecasting, causal inference/A-B testing, LLM-RAG pipelines) within large-scale pricing optimization systems. Led cross-functional Agile squads, mentored junior scientists, and communicated model performance and trade-offs to technical and business stakeholders.

- Built a custom clustering algorithm integrated into the core pricing engine, improving price recommendation accuracy in 80% of scenarios compared to baseline models.
- Enhanced time-series forecasting pipelines (Python, Prophet, scikit-learn, MLflow), leading to higher reliability and reduced operational intervention.

Senior Data Scientist Oct 2022 – Oct 2024 | Barcelona

Sales and Marketing – Ricoh

Designed, developed, and deployed predictive analytics and ML-powered tools to support sales planning, marketing campaigns, performance tracking, and strategic account management. Collaborated closely with product, sales, and marketing leaders to embed insights into operational workflows.

- Built predictive models for customer prioritization and territory optimization, influencing sales rep action planning and account management strategy
- Delivered deep-dive segmentation and behavioral insights, informing new commercial playbooks such as optimizing calls vs. in-person visits by client value tier.
- Developed automated reporting and dashboard systems that **reduced manual reporting effort** and improved decision-making transparency across regions.

Data Scientist Aug 2020 – Sep 2021 | Bogotá

Financial Area – Superintendencia Nacional de Salud

Applied ML and statistical analysis to address financial challenges including forecasting, tariff evaluation, and stakeholder segmentation. Simulated revenue and collection trends across economic scenarios and designed dashboards for policy-making support.

- Developed ML models to forecast collections under various tariff schemes, improving financial planning accuracy and resource allocation.
- Created interactive dashboards in Power BI, uncovering trends in financial data and enhancing decision-making across departments.
- Designed a new tariff scheme combining economics and ML, resulting in a 50%+ increase in revenue from control levies.

Data Scientist – Team Leader Aug 2017 – Feb 2020 | Bogotá

Collection Management – Secretaría de Movilidad

Led analytics for collection management services. Designed data pipelines, performance KPIs, and machine learning tools to predict risk and enhance budget allocation. Played a key role in modernizing and scaling operational processes.

- Improved six major operational workflows (covering 90% of the department) by introducing a rule-based clustering system for legal case classification.
- Created an ML model to assess default risk and optimize intervention strategies, helping allocate limited resources more effectively and reduce revenue leakage.

Economist

Jan 2016 – Dec 2017 | Bogotá

GEIH Survey – DANE

Analyzed city-level labor market trends and published monthly statistical bulletins. Retrieved and managed data using SQL and developed reports and presentations to support policymaking and public dissemination.

- Proposed two national labor market reports—one on ethnic minorities and one on older populations—filling critical data gaps in national statistics.
- Optimized SQL queries used to retrieve labor and social indicators, significantly reducing processing time and improving data responsiveness for stakeholders.

Other work experiences: Projects developer (1 year), Graduate Research Assistant (2 years) and Market Analysis (1 year)

EDUCATION	
M.S in Data Science	2021-2022
Barcelona School of Economics, Barcelona, Spain	
Postgraduate Studies in Statistics	2016-2017
Universidad Nacional de Colombia, Bogotá, Colombia	
Economics	2008-2012
Universidad Militar Nueva Granada, Bogotá, Colombia	

TECH STACK	
Backend development:	Python, FastAPI
Machine learning and AI frameworks:	Scikit-learn, TensorFlow, LangChain, Torch, XGBoost, MLFlow
Cloud computing:	AWS (Lambda, Athena, ECR, DynamoDB, Glue), Airflow
Data Base / BIG Data	SQL, PySpark
Deployment	Docker
Statistical Software:	R, SAS, Stata
Visualization/Reporting Tools:	Power BI, Looker, Tableau
Other Skills:	GIT, Agile Methodologies, MLOps